

Technology #12

Osmotic Pressure Regulating Drainage Valve

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Osmotic Pressure Regulating Drainage Valve

WHY DOES IT MATTER?

Fixed-pressure valves cannot adapt to changing CSF conditions

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 11 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Osmotic Membrane Transport + Membrane Mechanics. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Neuroshunt manufacturer; membrane technology company.

WHAT DOES THE BUYER GET?

Osmotic regulation concept + mechanism description; Van 't Hoff + Kedem-Katchalsky governing equations; Bench test protocols (permeation, fouling, aging, drainage); External precedent catalog (osmotic pumps: DURECT, Alzet); Critical UNKNOWN disclosure (membrane fouling rate, reservoir depletion rate); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production membrane design (material, geometry, L_p , σ); Osmotic reservoir design (agent, volume, lifetime); Manufacturing process for membrane + reservoir + valve integration; Regulatory submission (likely Class III PMA — no predicate for osmotic CSF valve); Clinical validation (drainage performance endpoint); Long-term fouling + depletion data

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Membrane specimens (various materials + geometries)

WHAT WOULD MAKE US KILL IT?

Membrane fouling rate in CSF reduces L_p by >50% within 30 days (insufficient for chronic implant)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Membrane specimens (various materials + geometries)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Membrane specimens (various materials + geometries) - 10 weeks

Basis: Build plan WP. Scope: Lp, sigma, burst pressure. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Fouling test articles (membrane in CSF mimic with protein) - 12-24 weeks (chronic)

Basis: Build plan WP. Scope: Lp vs time with protein challenge. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens (PES membrane + osmotic reservoir housing + NaCl agent) - 12 weeks (external)

Basis: Build plan WP. Scope: ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: Membrane is ideally semipermeable ($\sigma = 1$) or partially permeable ($\sigma < 1$); Osmotic agent concentration stable over implantation lifetime (UNKNOWN — depends on reservoir design); CSF protein concentration does not foul membrane significantly (UNKNOWN); Temperature constant at 37°C; Membrane properties (L_p , σ) stable over lifetime (UNKNOWN). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Membrane specimens (various materials + geometries). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Semipermeable membrane with osmotic reservoir provides self-regulating drainage based on combined hydrostatic and osmotic pressure

POTENTIAL VALUE DRIVER: Addresses: Fixed-pressure valves cannot adapt to changing CSF conditions

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Neuroshunt manufacturer; membrane technology company

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Osmotic Membrane Transport + Membrane Mechanics

Disciplines: Membrane science (semipermeable membranes), Osmotic transport (van 't Hoff, Kedem-Katchalsky), Biomaterials (membrane fouling resistance), Chemical engineering (solute transport)

Architecture: CSF shunt valve regulated by osmotic pressure differential across semipermeable membrane. Osmotic agent concentration on one side of membrane modulates effective drainage pressure based on CSF protein concentration or implanted osmotic reservoir.

Subsystems:

SS-01: Semipermeable membrane - Selectively permeable to water, retains solutes; creates osmotic pressure differential

SS-02: Osmotic agent reservoir - Contains solute at fixed concentration; provides osmotic pressure reference

SS-03: CSF-facing chamber - CSF contacts membrane; osmotic differential drives water flux across membrane

SS-04: Drainage outlet - Water (or CSF) drains to peritoneum based on net osmotic + hydrostatic pressure

2. Mechanism Architecture

Physical Changes: CSF hydrostatic pressure + osmotic pressure differential across semipermeable membrane drive water flux; osmotic reservoir maintains reference pressure

Key Physics: Two pressure sources in opposition: hydrostatic (ICP - outlet pressure) and osmotic (proportional to solute concentration differential). Net flux $J_v = L_p * (dP_{\text{hydrostatic}} - \sigma * dP_{\text{osmotic}})$.

3. Governing Engineering Model

Summary: Semipermeable membrane flux governed by Kedem-Katchalsky equations. Net water flux depends on hydraulic pressure differential and osmotic pressure differential, scaled by membrane permeability L_p and reflection coefficient σ .

Equations:

$\pi = i * C * R * T$ [osmotic pressure]

$J_v = L_p * (dP - \sigma * d\pi)$ [Starling/Kedem-Katchalsky]

$J_v = L_p * (dP - \sigma * d\pi)$

$Q_{\text{drainage}} = J_v * A_{\text{membrane}}$ [volumetric drainage rate]

$\pi = i * C * R * T$ [van 't Hoff osmotic pressure]

Assumptions:

- Membrane is ideally semipermeable ($\sigma = 1$) or partially permeable ($\sigma < 1$)
- Osmotic agent concentration stable over implantation lifetime (UNKNOWN — depends on reservoir design)
- CSF protein concentration does not foul membrane significantly (UNKNOWN)
- Temperature constant at 37°C
- Membrane properties (L_p , σ) stable over lifetime (UNKNOWN)

Boundary Conditions:

- CSF side: ICP 5-40 mmHg + CSF osmotic pressure (~290 mOsm/kg physiological)
- Reservoir side: hydrostatic pressure + osmotic agent concentration (design-dependent)
- Membrane area A_{membrane} (design-dependent)
- Temperature: 37°C constant

Failure Regimes:

- L_p too high -> excessive drainage (over-drainage)
- L_p too low -> insufficient drainage (under-drainage)

- $\sigma < 1$ (leaky membrane) $\rightarrow$ osmotic agent leaks, regulation lost
- Membrane fouling $\rightarrow$ L_p decreases over time, drainage drops
- Osmotic reservoir depletion $\rightarrow$ dP_i drops, drainage increases (potentially over-drainage)

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Passive self-regulating CSF drainage; existing valves are fixed-pressure or pressure-activated (R332)	VERIFIED
DI-002	Functional requirement	Osmotic pressure regulating drainage valve using semipermeable membrane	MODELLED
DI-003	CSF osmolarity (physiological)	EXTERNAL_PRECEDENT (~290 mOsm/kg)	EXTERNAL_PRECEDENT
DI-004	ICP range	ICP 5-40 mmHg (osmotic regulation range; membrane flux responds to combined hydrostatic + osmotic)	VERIFIED
DI-005	Drainage target	MODELLED ~0.3 mL/min typical	MODELLED
DI-006	Membrane properties (Lp, sigma)	UNKNOWN — design choice	UNKNOWN
DI-007	Osmotic agent	UNKNOWN — design choice (NaCl, mannitol, dextran, etc.)	UNKNOWN
DI-008	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-009	Sterilization	UNKNOWN — membrane may be gamma-sensitive; aseptic or EtO	UNKNOWN
DI-010	Membrane fouling environment	CSF contains proteins, cells — fouling risk	EXTERNAL_PRECEDENT
DI-011	ISO 7197 (CSF shunts)	UNKNOWN — applicability to be confirmed	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Semipermeable membrane design (material, geometry, sigma)	ABSENT	Membrane material selection; Lp target; sigma target; Geometry
DO-002	Osmotic reservoir design (volume, agent, replenishment)	ABSENT	Agent selection; Reservoir volume; Lifetime target
DO-003	Integrated valve + membrane + reservoir assembly	ABSENT	Component layout; Housing design
DO-004	Bench test protocol (permeation, fouling, aging)	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Membrane hydraulic permeability	UNKNOWN — design choice	m/(Pa·s)	Membrane selection	ASTM D3985 permeation test
Reflection coefficient sigma	UNKNOWN — design choice	dimensionless	Membrane selection	Membrane characterization
Osmotic agent concentration	UNKNOWN — design choice (e.g. NaCl, mannitol)	Design choice		
Membrane area A	UNKNOWN	m^2	Geometry	
Reservoir volume	UNKNOWN — determines lifetime before depletion	Design choice		
Membrane fouling rate	UNKNOWN in CSF environment	%/day	UNKNOWN	Fouling study in CSF mimic

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Membrane fouling	CSF proteins, cells, or debris	Fouling-resistant membrane	Fouling study in CSF mimic	UNKNOWN fouling rate in vivo
Osmotic reservoir depletion	Osmotic agent diffuses across membrane	Large reservoir, over-engineered membrane	Aging study with agent permeability	UNKNOWN depletion rate
Membrane rupture	Mechanical or chemical damage to membrane	High tensile strength material selection	Burst pressure test	UNKNOWN in vivo mechanical environment
Lp out of spec (too high)	Manufacturing variation or material defect	Acceptance tolerance analysis	Cpk measurement	UNKNOWN achievable Cpk
Osmotic agent biocompatibility	Osmotic agent leaks into CSF or dissolves in CSF	ISO 10993 biocompatibility testing	ISO 10993 (e.g. NaCl irritancy)	UNKNOWN long-term biocompatibility
Over-drainage if reservoir depletes	Reservoir depletes, dPi drops, drainage hydrostatic valve retracts	Reservoir depletion simulation	Reservoir depletion simulation	UNKNOWN failure rate
CSF osmolarity variation	Patient CSF osmolarity variation	Design for physiological range	Sensitivity analysis	UNKNOWN clinical CSF osmolarity range

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Membrane fouling (CRITICAL)	CSF proteins/cells accumulate on membrane	Fouling-resistant coating + larger area	UNKNOWN fouling rate in vivo
Osmotic reservoir depletion	Agent diffuses across membrane	Larger reservoir + low-permeability membrane	UNKNOWN depletion rate
Membrane rupture	Mechanical/chemical damage	Material selection + mechanical protection	UNKNOWN in vivo mechanical environment
Lp out of spec (manufacturing)	Manufacturing variation	Cpk study + tolerance	UNKNOWN achievable Cpk
Osmotic agent biocompatibility	Agent leaks into CSF/tissue	Biocompatible agent + low-leak membrane	UNKNOWN long-term biocompatibility
Over-drainage if reservoir fails	Reservoir depletion -> dPi drops -> drainage	Secondary safety:rostatic valve as safety flow	UNKNOWN failure rate
CSF osmolarity variation	Patient CSF osmolarity varies	Margin in design	UNKNOWN clinical osmolarity range

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Membrane Lp and sigma within target	ASTM D3985	Within target	NOT_TESTED
V-002	Drainage rate in physiological range	Bench flow	Q in target range	NOT_TESTED
V-003	Self-regulation (pressure response)	Pressure step bench test	Proportional response	NOT_TESTED
V-004	Membrane stability over lifetime	Aging with Lp measurement	Drift < 30%	NOT_TESTED
V-005	Fouling resistance	Protein challenge study	Lp reduction < 30%	NOT_TESTED
V-006	Biocompatibility ISO 10993	ISO 10993	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Drainage regulation in animal model	In vivo (IACUC)	Drainage within safe range	NOT_PERFORMED
VAL-002	Clinical drainage performance	Clinical trial (IDE)	Comparable or superior to standard	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Semipermeable membrane	Cellulose acetate	Standard dialysis membrane material	ISO 10993 + Lp/sigma characterization	CANDIDATE
Semipermeable membrane	Polyethersulfone (PES)	Standard membrane material	ISO 10993 + Lp/sigma + fouling	CANDIDATE
Semipermeable membrane	Polyamide (nylon)	Membrane material	ISO 10993 + Lp/sigma + fouling	CANDIDATE
Osmotic agent	NaCl (sodium chloride)	Standard physiological osmotic agent	Purity + leachability + biocompatibility	CANDIDATE
Osmotic agent	Mannitol	Standard osmotic agent in medical use	Purity + leachability	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Semipermeable membrane (custom)	1	CUSTOM_COMPONENT	UNKNOWN — likely cellulose acetate, PES, or nylon	UNKNOWN	CRITICAL	Purity + sigma + biocompatibility
02	Osmotic reservoir housing	1	CUSTOM_COMPONENT	UNKNOWN — likely polyimide or polyethylene	UNKNOWN	CRITICAL	Hermeticity + biocompatibility
03	Osmotic agent (NaCl, mannitol, or dextrose)	Reservoir fillable	CONTAINABLE	NaCl or mannitol or dextrose (USP-grade)	Multiple	MEDIUM	Purity + leachability
04	Valve housing + catheter connection	1	CUSTOM_COMPONENT	UNKNOWN — likely stainless steel or Ti composite	Multiple	MEDIUM	Dimensional + biocompatibility

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no osmotic CSF valve process validated

Candidate Processes:

- Membrane fabrication (phase inversion or track-etch): Standard membrane manufacturing - Requires membrane manufacturer partnership
- Reservoir filling + sealing: Standard osmotic pump manufacturing (DURECT) - Aseptic filling required if terminal sterilization incompatible
- Housing injection molding + assembly: Standard implantable device manufacturing - Custom for membrane integration

14. External Evidence

Source 1: Cerebral Spinal Fluid (CSF) Shunt Systems - FDA

URL: <https://www.fda.gov/medical-devices/implants-and-prosthetics/cerebral-spinal-fluid-csf-shunt-systems>

Excerpt: Jul 16, 2018 · Fixed shunt valves allow CSF fluid to drain when CSF pressure exceeds a certain “fixed” threshold. Adjustable shunt valves allow for changes to ...

Source 2: Understanding Shunt Systems | Hydrocephalus Association

URL: <https://www.hydroassoc.org/shunt-systems>

Excerpt: A shunt is a flexible tube placed into the area of the brain called the ventricles, where cerebrospinal fluid (CSF) is accumulating.

Source 3: CSF-Flow Control Valves - Shunt for Hydrocephalus - Medtronic

URL: <https://www.medtronic.com/en-us/l/patients/treatments-therapies/hydrocephalus-shunt/csf-flow-control-valves.html>

Excerpt: A shunt redirects excess cerebrospinal fluid (CSF) in the brain to another part of the body, allowing the brain's enlarged ventricles to return to a more normal ...

Source 4: A Review of Cerebral Shunts, Current Technologies, and Future Endeavors

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6153620>

Excerpt: Sep 21, 2018 · The valve is responsible for the release of CSF from the brain into the second catheter that drains the CSF to the absorption site.

Source 5: Brain shunt - Mayo Clinic

URL: <https://www.mayoclinic.org/tests-procedures/brain-shunt/about/pac-20588547>

Excerpt: Sep 5, 2025 · The valve controls the rate of CSF flow from the brain to the drainage site. This allows the fluid to be absorbed elsewhere and relieves ...

15. Transfer Boundary

YOU RECEIVE:

- + Osmotic regulation concept + mechanism description
- + Van 't Hoff + Kedem-Katchalsky governing equations
- + Bench test protocols (permeation, fouling, aging, drainage)
- + External precedent catalog (osmotic pumps: DURECT, Alzet)
- + Critical UNKNOWN disclosure (membrane fouling rate, reservoir depletion rate)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production membrane design (material, geometry, Lp, sigma)
- > Osmotic reservoir design (agent, volume, lifetime)
- > Manufacturing process for membrane + reservoir + valve integration
- > Regulatory submission (likely Class III PMA — no predicate for osmotic CSF valve)
- > Clinical validation (drainage performance endpoint)
- > Long-term fouling + depletion data

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable membrane Lp and sigma in CSF environment — UNKNOWN
- Membrane fouling rate in CSF — UNKNOWN (critical)
- Osmotic reservoir depletion rate — UNKNOWN
- Optimal osmotic agent — UNKNOWN (NaCl vs mannitol vs dextran, etc.)
- Membrane lifetime in vivo — UNKNOWN
- Regulatory pathway (likely Class III PMA — no predicate for osmotic CSF valve) — UNKNOWN
- Clinical benefit magnitude for osmotic regulation — drainage stability vs fixed-pressure valve across CSF osmolarity variation — UNKNOWN until trial

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Membrane specimens (various materials) (ASTM D2995)	Permeation test cell (AP01100235)	Pressure + resistance	Target range	Age band spec	10 weeks
WP-02	Integrated valve prototype (inflow loop, no CSF)	2665g pressure source	Flow rate, target regulation	Optimal response	40 weeks	
WP-03	Fouling test articles (mainly chamber tips with protein)	CSF, protein challenge	Protein challenge < 30%	Fouling rate	12-24 weeks (chronic)	
WP-04	Reservoir aging specimens	Aging chamber, osmotic agent	Osmotic agent concentration	Retention	Res 50% stability	12-24 weeks
WP-05	Biocompatibility specimens (ISO 10993)	Membrane + reservoir	Passing + NaCl agent	ISO 10993 report	12 weeks (external)	

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-06	In vivo drainage study	Animal facility	Drainage regulation	Drainage model in safe range	In vivo drainage study	2 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Neuroshunt manufacturer; membrane technology company.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Membrane fouling rate in CSF reduces L_p by >50% within 30 days (insufficient for chronic implant)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Membrane fouling rate in CSF reduces L_p by >50% within 30 days (insufficient for chronic implant)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.